

Complete Detection Comparison
***Seg-to-Det requires no detection-specific training (uses segmentation output)**

Model	Type	Paradigm	Params	AP@0.5	AP@50-95	Precision	Recall	Train Time
RT-DETR-L	Single-Task	Direct Det	32.0M	87.95%	57.38%	88.15%	89.38%	34 min
YOLOv8m	Single-Task	Direct Det	25.9M	91.22%	59.56%	87.45%	88.47%	30 min
WS-MINI (Seg-to-Det)	Multi-Task	Seg-to-Det	34.3M	86.70%	-	-	-	N/A*
WS-BASE (Seg-to-Det)	Multi-Task	Seg-to-Det	520.4M	89.91%	-	-	-	N/A*
WS-XL (Seg-to-Det)	Multi-Task	Seg-to-Det	762.5M	96.23%	-	96.23%	94.64%	N/A*